

MUTUAL FUND ANALYTICS

End-to-End Data Analytics, Performance & Risk Assessment of Indian Mutual Funds

FINAL PROJECT REPORT

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Project Domain: Data Analytics / Financial
Analytics

Technologies Used:

Python | Pandas | SQL | SQLite | Excel | Power BI

Academic Year: 2026–27

PAGE 2 — EXECUTIVE SUMMARY

1. Executive Summary

The Mutual Fund Analytics project is an end-to-end data analytics solution designed to analyse mutual fund industry trends, scheme performance, investor behaviour, SIP growth and investment risk. The project integrates multiple datasets containing fund master information, historical NAV, assets under management, SIP inflows, category inflows, investor transactions, portfolio holdings and benchmark indices.

The primary objective of the project is to transform raw and heterogeneous mutual fund data into structured analytical datasets and meaningful business insights. Python and Pandas are used for data cleaning and transformation, SQLite is used for analytical data storage, SQL is used for structured querying, and Power BI is used for interactive visualization and dashboard development.

The project covers 40 mutual fund schemes and calculates several financial performance and risk metrics including daily returns, CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown, Historical VaR and CVaR. A composite fund scorecard is also developed using return, risk-adjusted performance, alpha, expense ratio and maximum drawdown.

The analysis indicates strong growth in SIP activity. Monthly SIP inflow increases from approximately ₹11,517 crore in January 2022 to ₹31,002 crore in December 2025. Active SIP accounts also show significant growth, increasing from approximately 4.91 crore to 9.35 crore.

The composite scorecard demonstrates that fund selection should not be based only on historical return. Risk-adjusted performance, benchmark-relative performance, expenses and downside risk are important factors when comparing mutual fund schemes.

PAGE 3 — PROBLEM & OBJECTIVES

2. Problem Statement

Mutual fund data is distributed across multiple datasets containing scheme information, NAV history, AUM, SIP inflows, investor transactions, portfolio holdings and benchmark indices. Analysing these datasets independently makes it difficult to obtain a unified view of fund performance, risk and investor behaviour.

The project addresses this problem by developing an integrated analytics pipeline that cleans, transforms, stores and analyses the datasets and presents the results through an interactive Power BI dashboard.

2.1 Objectives

- To collect and organize mutual fund datasets.
- To clean and preprocess raw datasets.
- To standardize dates, identifiers and numerical fields.
- To analyse AUM and SIP trends.
- To study investor transaction behaviour.
- To analyse category-wise inflows.
- To calculate fund performance metrics.
- To calculate risk metrics.
- To compare funds with relevant benchmarks.
- To develop a composite fund scorecard.
- To create an interactive Power BI dashboard.

- **PAGE 4 — DATA SOURCES**

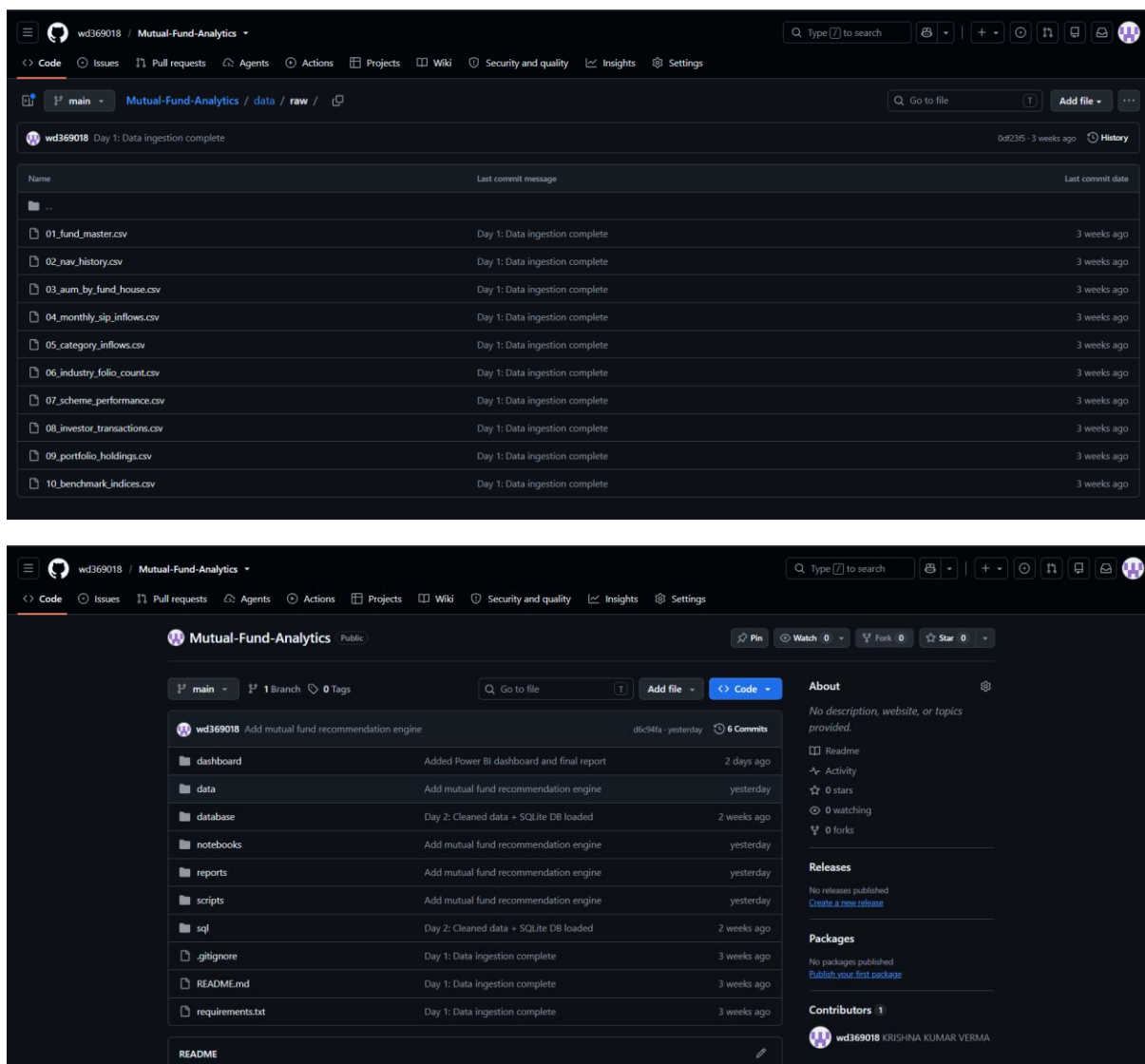
- **3. Data Sources**

- The project uses multiple structured datasets covering mutual fund schemes, NAV history, AUM, SIP activity, investor transactions, portfolio holdings and benchmark indices.

Dataset	Purpose
01_fund_master	Scheme and fund information
02_nav_history	Historical NAV data
03_aum_by_fund_house	AMC-level AUM analysis
04_monthly_sip_inflows	Monthly SIP trends
05_category_inflows	Category-wise inflow analysis
06_industry_folio_count	Folio and investor participation
07_scheme_performance	Scheme performance
08_investor_transactions	Investor behaviour
09_portfolio_holdings	Portfolio exposure
10_benchmark_indices	Benchmark comparison

Data Characteristics

The analysis universe contains **40 mutual fund schemes**. Historical NAV data is used for calculating daily returns and risk metrics. Benchmark data contains indices such as NIFTY50, NIFTY100, NIFTY500, NIFTY Midcap 150, CRISIL Liquid and CRISIL Gilt.



Caption:

Figure 3.1: Project data repository and dataset structure

PAGE 5 — ETL DESIGN

4. ETL Design

The project follows an Extract, Transform and Load architecture.

Extract

Raw CSV files are collected and loaded into the Python environment. The datasets contain information related to mutual funds, NAV, AUM, SIP, investor transactions and benchmarks.

Transform

The transformation stage includes:

- Data type conversion
- Date parsing
- Sorting
- Missing-value handling
- Duplicate removal
- Identifier standardization
- Numerical conversion
- Return calculation
- Derived metric creation

Load

The cleaned datasets are stored in processed form and loaded into SQLite for analytical querying.

MUTUAL FUND DATA SOURCES

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Raw CSV Files
Fund Master
NAV History
AUM Data
SIP Inflows
Transactions
Performance
Portfolio Holdings
Benchmark Indices

|



ETL PROCESS
Extract → Transform
- Data Cleaning
- Missing Values
- Date Formatting
- Duplicate Removal
- Data Validation
→ Load

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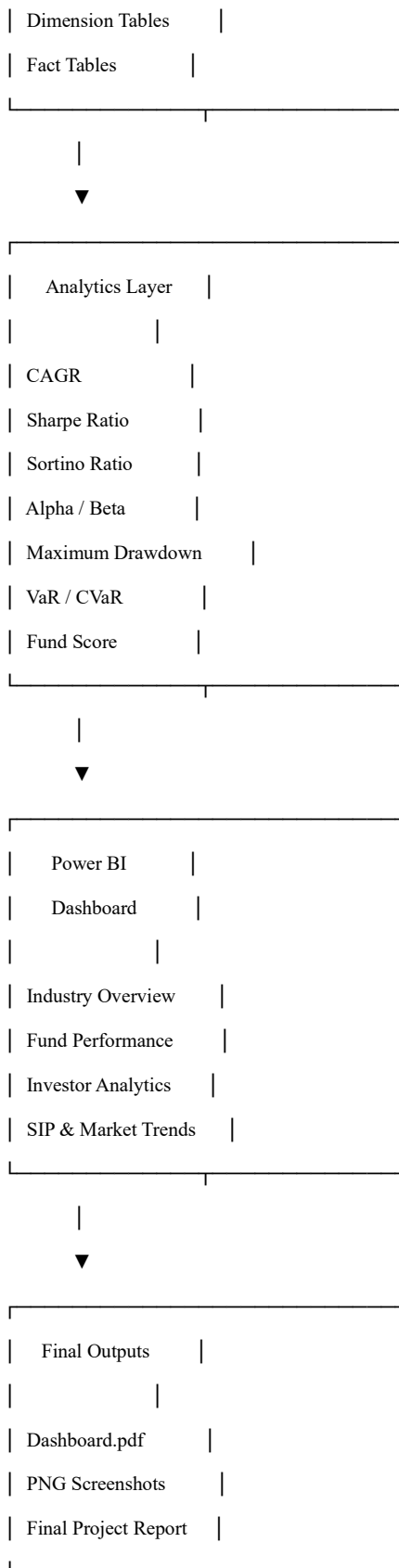


Processed / Clean Data
CSV Files

|



SQLite Database



PAGE 6 — DATABASE & ARCHITECTURE

5. Analytical Architecture

The project is divided into five major layers:

1. **Source Layer** – Raw CSV and benchmark datasets.
2. **Processing Layer** – Python and Pandas based data cleaning.
3. **Storage Layer** – SQLite analytical database.
4. **Analytics Layer** – Python notebooks and SQL queries.
5. **Presentation Layer** – Power BI dashboard and final report.

The database design follows a dimensional approach with fund and date dimensions and fact tables for NAV, performance, transactions and AUM.

AMFI code and date act as important integration fields between datasets.

The architecture provides separation between raw data, processed data, analytical calculations and visualization, making the project easier to maintain and reproduce.

6. Exploratory Data Analysis

6.1 AUM Analysis

AUM analysis was performed to understand the growth of assets managed by major mutual fund houses.

The project data shows significant growth:

Fund House	Earlier AUM	Later AUM
SBI	₹6.05 lakh crore	₹12.50 lakh crore
ICICI Prudential	₹4.65 lakh crore	₹10.74 lakh crore
HDFC	₹4.35 lakh crore	₹9.30 lakh crore

Key Findings

- SBI remained one of the largest fund houses in the selected dataset.
- Major fund houses showed strong AUM growth.
- AUM provides an important indicator of scale.
- AUM should not be treated as a direct indicator of investment quality.

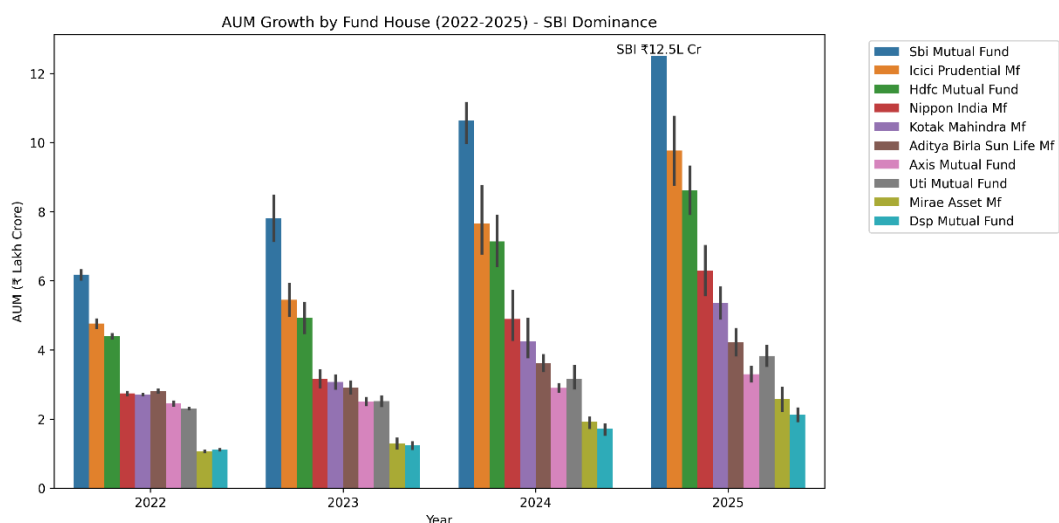


Figure 6.1: AUM growth of major fund houses

PAGE 8 — SIP ANALYSIS

6.2 SIP Inflow Analysis

SIP analysis is an important component of the project because systematic investment is a major indicator of retail investor participation.

The dataset shows monthly SIP inflow increasing from approximately:

₹11,517 crore → ₹31,002 crore

between January 2022 and December 2025.

Active SIP accounts also increased approximately from:

4.91 crore → 9.35 crore

This indicates increasing participation in systematic investment.

Key Findings

- SIP inflows increased substantially.
- Active SIP accounts also increased.
- Growth was supported by increasing investor participation.

Monthly SIP Inflow Trend (Jan 2022 - Dec 2025)

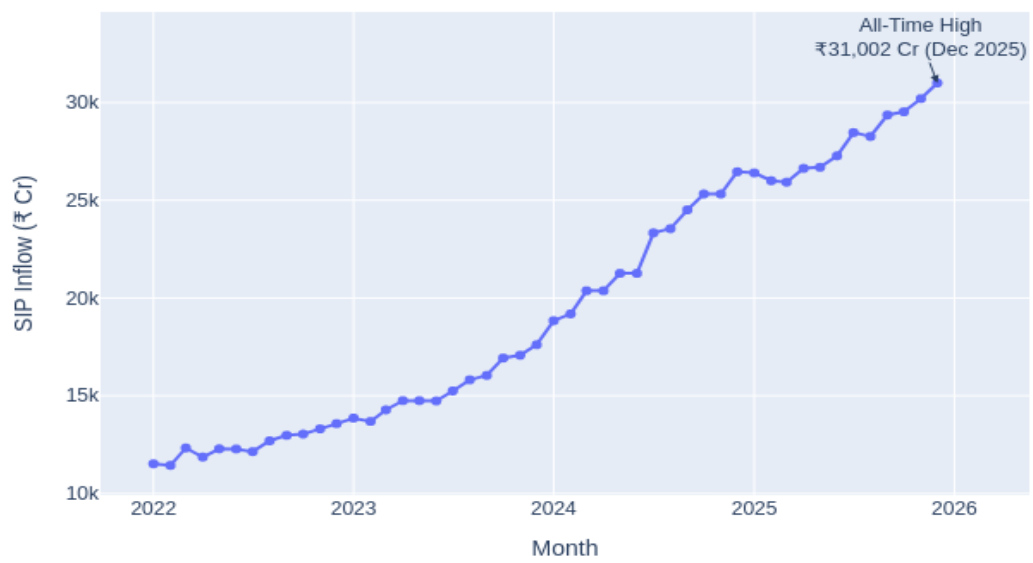


Figure 6.2: Monthly SIP inflow trend

PAGE 9 — INVESTOR ANALYTICS

6.3 Investor Behaviour Analysis

Investor transaction data was analysed using transaction type, state, age group and cohort information.

The dashboard allows users to examine:

- SIP transactions
- Lumpsum transactions
- Redemptions
- State-wise investment activity
- Age-group behaviour
- Monthly transaction volume

Cohort Analysis

Cohort Average SIP

2024 ₹10,996.89

2025 ₹13,505.21

The 2025 cohort shows a higher average SIP amount. However, because the number of observations differs significantly between the cohorts, this result should be interpreted as a directional observation rather than a generalized population-level conclusion.

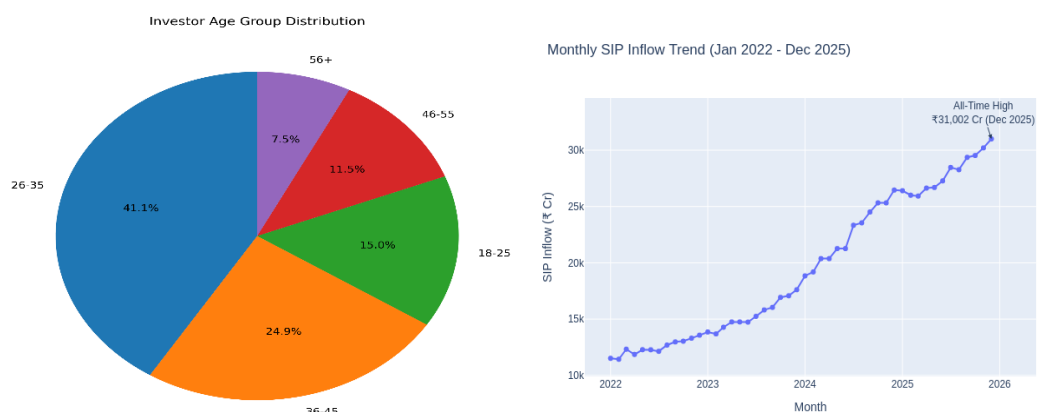


Figure 6.3: Investor Analytics Dashboard

PAGE 10 — CATEGORY ANALYSIS

6.4 Category-wise Inflow Analysis

The project contains category-level inflow data covering categories such as:

- Large Cap
- Mid Cap
- Small Cap
- Flexi Cap
- Large & Mid Cap
- ELSS
- Value/Contra
- Sectoral/Thematic
- Liquid
- Short Duration
- Gilt
- Hybrid

Category inflow analysis helps identify where investors are allocating capital.

However, higher inflow does not automatically indicate higher future performance. Investment flows and investment returns represent different dimensions of fund behaviour.

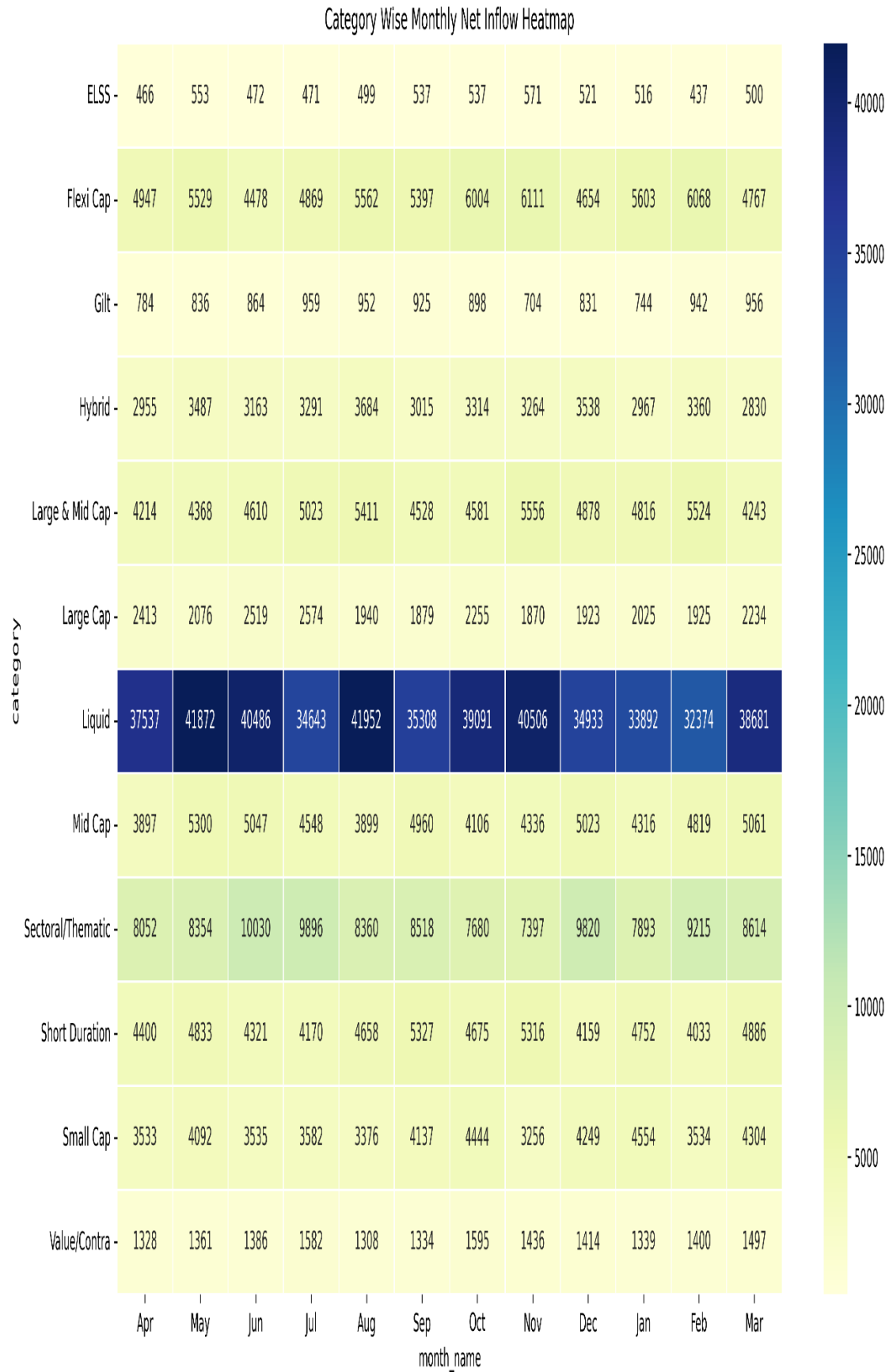


Figure 6.4: Category-wise mutual fund inflows

PAGE 11 — PERFORMANCE METHODOLOGY

7. Performance Analysis

The project uses multiple metrics to evaluate mutual fund performance.

Metric	Purpose
Daily Return	Measures daily NAV movement
CAGR	Measures annualized growth
Sharpe Ratio	Measures return relative to total risk
Sortino Ratio	Measures return relative to downside risk
Alpha	Measures benchmark-relative excess performance
Beta	Measures sensitivity to benchmark
Maximum Drawdown	Measures peak-to-trough decline
VaR	Measures historical loss threshold
CVaR	Measures average loss beyond VaR

A **6.5% risk-free-rate proxy** is used for Sharpe and Sortino calculations.

Benchmark returns are used for Alpha and Beta calculations.

PAGE 12 — FUND SCORECARD

8. Composite Fund Scorecard

The project uses a weighted composite score to compare mutual fund schemes.

Score Formula

30% × 3-Year Return Rank

•

25% × Sharpe Rank

•

20% × Alpha Rank

•

15% × Expense Ratio Rank

•

10% × Maximum Drawdown Rank

Top Funds

Rank	Fund	Score
1	SBI Small Cap Direct	72.75
2	Kotak Flexicap Regular	71.50
3	Kotak Emerging Equity Regular	70.50

Rank	Fund	Score
4	ABSL Small Cap Regular	68.25
5	SBI Small Cap Regular	64.38

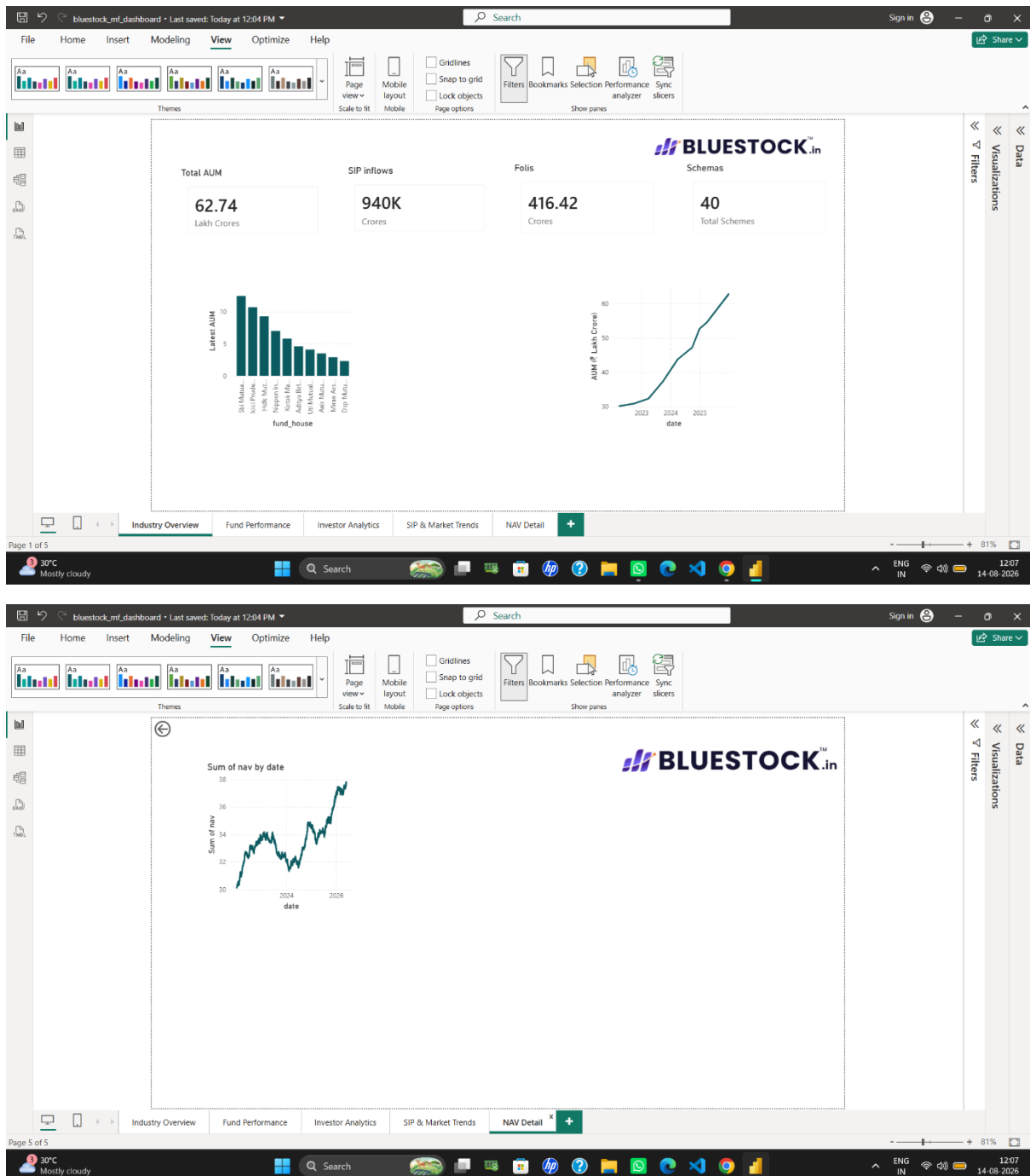


Figure 8.1: Top mutual funds by composite score

PAGE 13 — RISK ANALYSIS

9. Risk Analysis

9.1 Maximum Drawdown

Maximum drawdown represents the largest decline from a historical peak to the following trough.

Examples from the analysis:

- SBI Small Cap Direct: approximately **−52.57%**
- Axis Small Cap Regular: approximately **−51.68%**

These values demonstrate that high-return funds can also experience substantial downside periods.

9.2 Historical VaR

Historical 95% VaR is calculated as the **5th percentile of daily returns**.

9.3 CVaR

CVaR represents the average return among observations below the VaR threshold.

Example:

Scheme	VaR 95% CVaR 95%	
SBI Small Cap Direct	−2.69%	−3.24%
ABSL Small Cap Regular	−2.60%	−3.25%
Axis Small Cap Regular	−2.62%	−3.17%
Nippon India Small Cap	−2.54%	−3.23%
HDFC Mid-Cap Opportunities	−1.90%	−2.35%

10. Benchmark Analysis

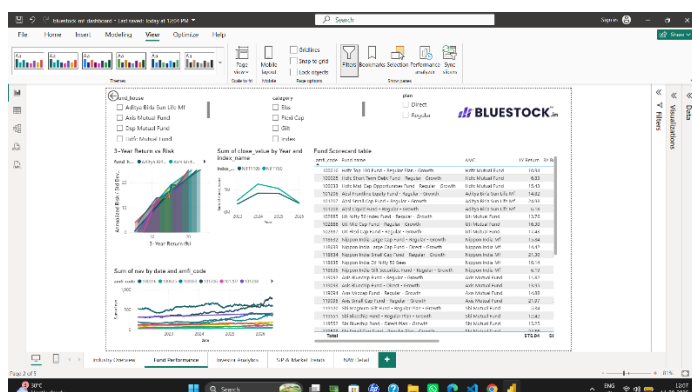
The project contains benchmark indices including:

- NIFTY50
- NIFTY100
- NIFTY500
- NIFTY Midcap 150
- CRISIL Liquid
- CRISIL Gilt
- BSE SmallCap

Alpha represents the excess performance of a fund relative to its benchmark after accounting for the regression relationship.

Beta measures the sensitivity of the fund to benchmark movements.

Benchmark analysis is important because a fund generating high absolute returns may not necessarily outperform its relevant benchmark.



PAGE 15 — POWER BI DASHBOARD

11. Dashboard Screenshots

The project includes a four-page Power BI dashboard.

Page 1 — Industry Overview

The dashboard contains:

- Total AUM
- SIP inflows
- Folios
- Schemes
- AUM trend
- AMC-wise AUM

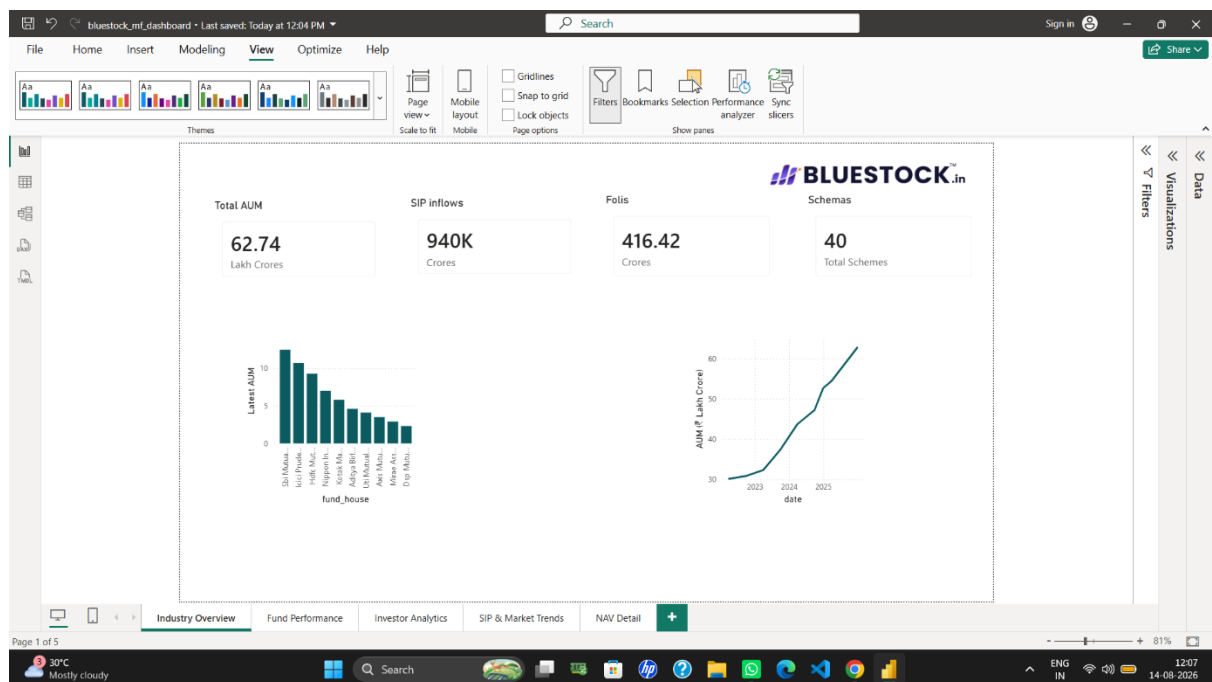


Figure 11.1: Industry Overview Dashboard

PAGE 16 — FUND PERFORMANCE

11.1 Fund Performance Dashboard

This page focuses on scheme-level performance.

Important visuals include:

- Return vs Risk scatter plot
- Fund scorecard
- NAV trend
- Benchmark comparison
- Risk indicators

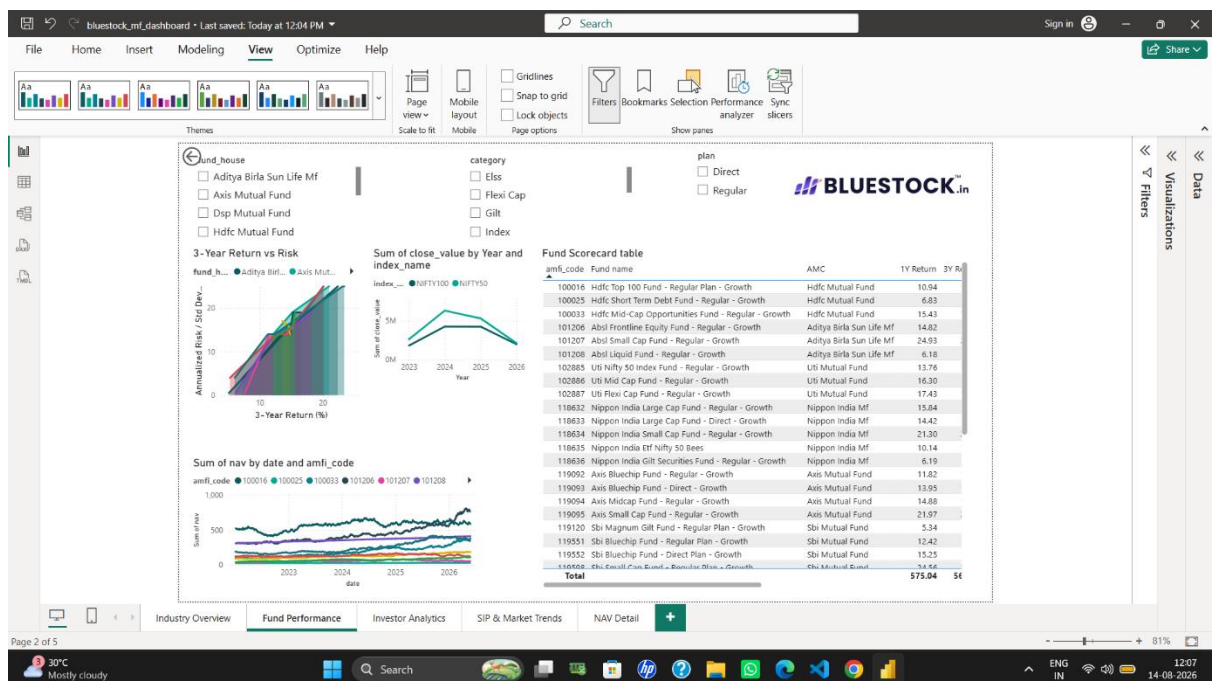


Figure 11.2: Fund Performance Dashboard

The page allows users to compare schemes using both return and risk measures instead of relying on a single performance metric.

PAGE 17 — INVESTOR ANALYTICS

11.2 Investor Analytics Dashboard

This page provides an investor-focused view of the dataset.

Key visualizations include:

- State-wise transaction amount
- Transaction-type distribution
- Age-group analysis
- SIP amount
- Monthly transaction volume

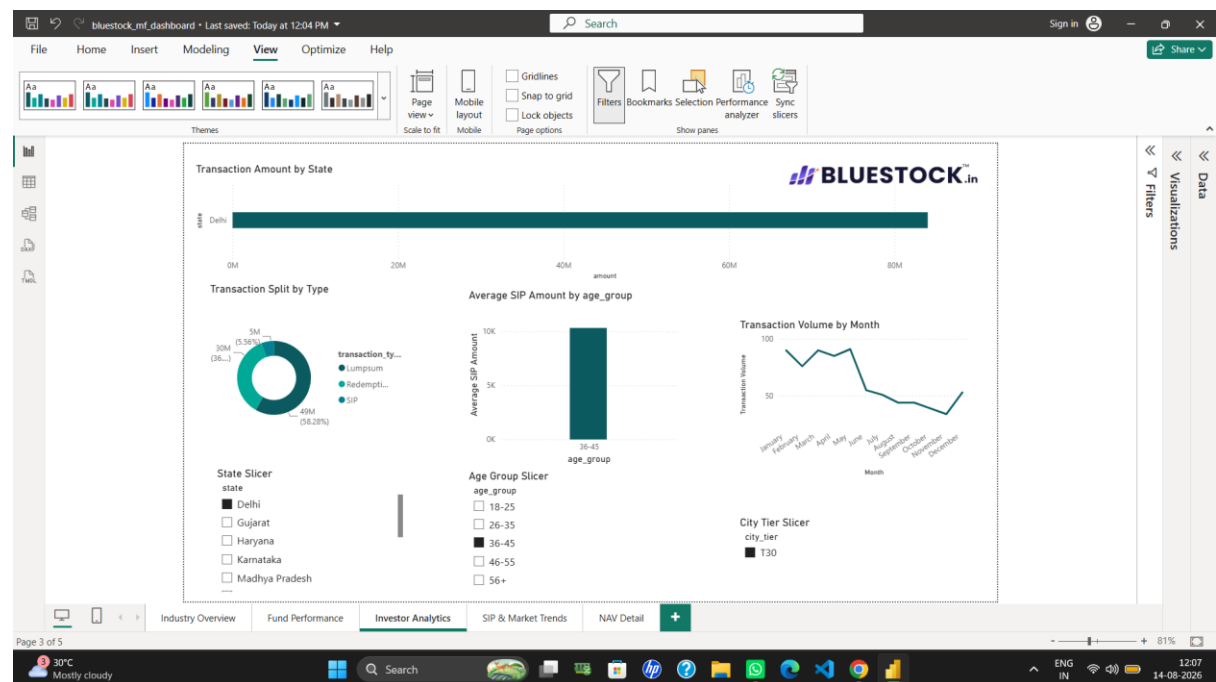


Figure 11.3: Investor Analytics Dashboard

PAGE 18 — SIP & MARKET TRENDS

11.3 SIP & Market Trends Dashboard

The fourth dashboard page combines SIP flows with market trends.

It contains:

- SIP inflow trend
- Market index trend
- Category inflow heatmap
- Top categories by net inflow

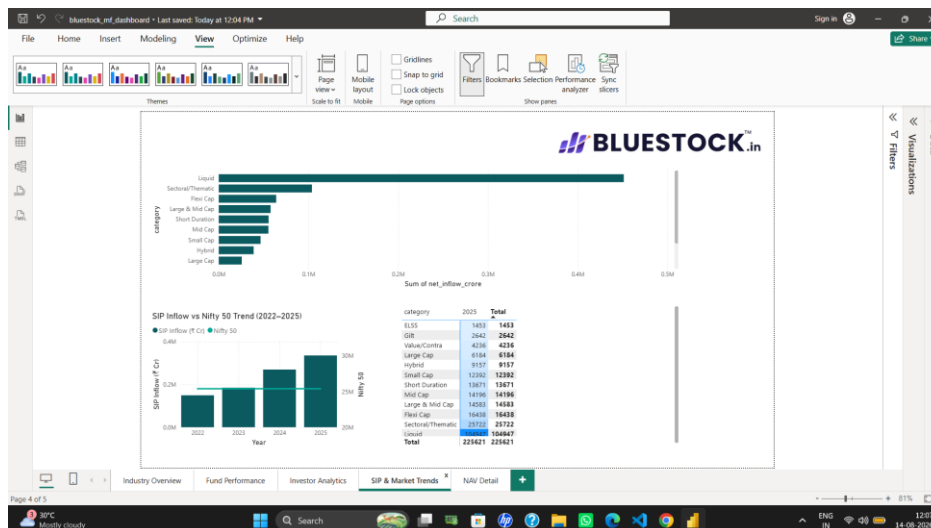


Figure 11.4: SIP and Market Trends Dashboard

This view helps analyse the relationship between investor flows and broader market movements.

PAGE 19 — LIMITATIONS & RECOMMENDATIONS

12. Limitations

- Analysis is limited to the schemes available in the project dataset.
- Historical performance does not guarantee future returns.
- VaR and CVaR depend on the selected historical period.
- Maximum drawdown is time-period dependent.
- Composite score weights are project-specific.
- Benchmark selection affects Alpha and Beta.
- Small investor cohorts can produce unstable averages.
- Live data may contain missing or delayed observations.
- Incorrect Power BI relationships can affect results.
- The dashboard is an analytics tool and not personalized investment advice.

13. Recommendations

1. Automate Data Refresh

Schedule regular data ingestion and validation.

2. Improve Benchmark Mapping

Use category-specific benchmarks.

3. Add Rolling Metrics

Add rolling:

- Sharpe
- Volatility
- Beta

- Drawdown

4. Improve Investor Analytics

Add SIP retention, redemption behaviour and investor lifetime value.

5. Improve Portfolio Analysis

Add sector concentration and portfolio overlap.

6. Improve Dashboard

Add:

- Last refresh date
 - Data-quality indicator
 - Drill-through
 - Advanced tooltips
-

PAGE 20 — CONCLUSION

14. Conclusion

The Mutual Fund Analytics project successfully demonstrates an end-to-end data analytics workflow involving data ingestion, cleaning, database design, SQL analysis, financial performance evaluation, risk analysis and Power BI visualization.

The project integrates multiple mutual fund datasets and converts raw information into meaningful insights related to AUM, SIP growth, investor behaviour, scheme performance and investment risk.

The analysis shows strong growth in SIP participation and AUM, while the performance analysis demonstrates that high returns can coexist with substantial downside risk. Therefore, mutual fund evaluation should consider return, risk-adjusted performance, benchmark-relative performance, expense ratio and drawdown together.

The composite fund scorecard provides a structured approach to comparing schemes. The project also demonstrates the practical integration of Python, Pandas, SQL, SQLite and Power BI in a financial analytics use case.

Future improvements can focus on automated data refresh, improved benchmark mapping, rolling risk metrics, portfolio concentration analysis and validation using larger and more recent datasets.

Overall, the project provides a strong foundation for a mutual fund research and analytics platform.

Project Repository

<https://github.com/wd369018/Mutual-Fund-Analytics/tree/main>